

Composition Pair B1.18 + B1.06: Content-Domain Specifications Require the DNA Layer as Their Authoritative Governed Substrate Home, and the DNA Layer's Governed Status Makes Content-Domain Specifications Authoritative Governing Content Inspectable and Modifiable Through Paper 1 Affordances

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize, in composition-pair form, the enabling relationship between content-domain (B1.18) and the two-layer structure (B1.06), demonstrating that neither concept is fully operational without the other and that their co-presence constitutes a necessary architectural unit.

Abstract

The Coordination Knowledge Substrate (CKS) Paper 2 architecture defines two foundational commitments that stand in enabling relationship to each other. B1.18 specifies that each entity has a content-domain — an authored operational territory defining what inputs the entity handles and what behaviors it performs, governed through substrate content. B1.06 specifies that every cell carries a two-layer structure: a DNA layer holding stabilized orchestration substrates, and an Action layer holding operational records. This note formalizes the enabling pair: content-domain specifications per B1.18 require the DNA layer per B1.06 as their authoritative governed substrate home, because without a separately-maintained, A1.08-authoritative substrate layer, content-domain specifications have no governed home and reduce to application configuration or documentation. Conversely, the DNA layer per B1.06 requires content-domain per B1.18 to hold operational territory specification alongside its behavior rules, because behavior rules govern how the entity operates while content-domain governs what territory the entity operates in. The note identifies the enabling direction, the significance direction, the mutual dependency structure, the distinct roles of both layers in content-domain governance, and two failure modes — Implicit Content-Domain (B3.19) and Layer Conflation (B3.07) — that follow when the pair is not co-present.

1. Pair Identification

Pair: B1.18 (content-domain) + B1.06 (two-layer structure)

Relationship type: Enabling pair

B1.18 — Content-domain names the authored operational territory of each entity: what input types are within the entity’s handling scope, what behaviors the entity performs when those inputs arrive, what inputs are out-of-domain, and how the entity responds to out-of-domain encounters. The content-domain commitment specifies that this territory is not assumed, not derived at runtime from inference, but explicitly authored as governed substrate content (B2.89–B2.93). Content-domain specifications include input domain schemas (B2.90), inclusion rules (B2.91), exclusion rules (B2.91), out-of-domain handling rules (B2.90), and boundary enforcement rules (B2.91).

B1.06 — Two-layer structure names the architectural division within every cell between the DNA layer and the Action layer. The DNA layer holds stabilized orchestration substrates and behavior substrates that define how the cell functions — the behavior rules, the conflict-handling rules, the lifecycle policies, the schemas. The Action layer holds recorded task instances and their outputs — what actually happened when the DNA met an actual task. Both layers are substrate content under Paper 1’s commitments; what distinguishes them is governance semantics. DNA is authored at origin and governed at modification; Action accumulates through operation and feeds back into DNA refinement through governed mechanisms (B2.25, B2.26).

Primary enabling direction: B1.06 enables B1.18 by providing the governed substrate layer in which content-domain specifications can reside as authoritative content. B1.18 enables B1.06 by providing the operational territory specification that gives the DNA layer’s behavior rules their governing scope.

2. The Enabling Direction: B1.06 Provides the Governed Substrate Home for B1.18

Content-domain per B1.18 is an authored commitment, not a runtime-inferred property. The architecture’s design requires that what inputs an entity handles, what inputs it refuses, and how it responds at the boundary are specified explicitly before the entity encounters any actual input — and that these specifications be authoritative substrate content subject to governance, not ad hoc configuration that may differ across deployments.

The DNA layer per B1.06 is what makes this possible. The DNA layer is the separately-maintained, A1.08-authoritative substrate layer within every cell. When content-domain specifications reside in the DNA layer, they acquire the substrate-as-source-of-truth property: they are what the system treats as the authoritative statement of the entity’s operational territory, not a copy, not a derivative, not an inferred reconstruction from operational behavior. This is what A2.46 Category 4 authoritative content means — the DNA layer’s content-domain specifications are the canonical record.

Four categories of content-domain specification reside in the DNA layer:

Input domain schemas (B2.90). These specify what types of inputs are in-domain for the entity — the structural, semantic, and contextual characteristics of inputs the entity is designed to handle. Input domain schemas are authored DNA-layer content; they define the entity’s operational territory in terms humans can inspect and modify.

Exclusion rules (B2.91). These specify what inputs are out-of-domain — the conditions under which an arriving input falls outside the entity’s handling scope. Exclusion rules are DNA-layer content, not runtime decisions; they are the authored policy the entity applies when evaluating incoming inputs.

Out-of-domain handling rules (B2.90). These specify how the entity responds to inputs it determines are out-of-domain: routing the input elsewhere, flagging for human attention, returning a governed response, or escalating per conflict-handling rules. Out-of-domain handling rules are DNA-layer content that governs the entity’s behavior at the boundary.

Boundary enforcement rules (B2.91). These specify how domain boundaries are enforced operationally — the mechanisms through which inclusion and exclusion determinations are made and applied. Boundary enforcement rules are DNA-layer content whose operational effect is to make the content-domain specification binding rather than advisory.

Without B1.06’s DNA layer as a separately-maintained, A1.08-authoritative substrate, content-domain specifications have nowhere governed to reside. They would exist as application configuration in whatever deployment environment the entity runs in, as documentation external to the substrate, or as implicit assumptions embedded in orchestration rules that do not make the territory explicit. Each of these alternatives degrades the content-domain commitment in a specific way: application configuration is not subject to Paper 1’s governance affordances; documentation external to the substrate is not A1.08-authoritative; implicit assumptions are not inspectable per A2.01 and not modifiable per A2.02 as architectural substrate. The DNA layer’s governed status is not incidental to content-domain; it is what transforms content-domain from design intent into authoritative architectural commitment.

3. The Significance Direction: B1.18 Provides Operational Territory Specification for the DNA Layer

The DNA layer per B1.06 holds behavior rules: the orchestration substrates governing how the entity functions, how it handles conflicts, how it executes its lifecycle, what its governance policies are. These behavior rules govern the entity’s operation once an input has arrived and been accepted. But behavior rules do not, by themselves, specify what the entity’s operational territory is — they govern behavior within territory, not the territory itself.

Content-domain per B1.18 is what provides that operational territory specification. Without B1.18, the DNA layer contains behavior specifications without scope: the entity has rules governing what it does, but the substrate does not specify what inputs should trigger those rules or what inputs fall outside the entity’s authority. This gap is not merely an organizational deficiency — it has operational

consequences. An entity without authored content-domain specifications cannot enforce its operational boundary from governed substrate content; enforcement must be rebuilt from whatever combination of behavior rules, application logic, or convention happens to be in place.

Content-domain's contribution to the DNA layer is specific and non-redundant with behavior rules. Behavior rules ask: given that this input is in-domain, how does the entity process it? Content-domain specifications ask: is this input in-domain? Together, they constitute complete operational specification. The DNA layer without content-domain is incomplete in a way that cannot be resolved by elaborating the behavior rules further — no amount of specification of how the entity processes in-domain inputs answers the question of what the in-domain territory is.

This is why B1.18 and B1.06 are a composition pair rather than merely adjacent commitments. B1.06's DNA layer requires B1.18's content-domain to be complete; B1.18's content-domain requires B1.06's DNA layer to be authoritative. Each commitment depends on the other for a property the other uniquely provides.

4. Mutual Dependency and Co-Presence Requirement

The mutual dependency in this pair has two faces.

B1.06 provides B1.18 governance infrastructure. The DNA layer per B1.06 is the substrate through which B1.18 content-domain receives its full governance treatment. The inspect right per A2.01 applies to content-domain specifications in the DNA layer: a human with appropriate access can read the input domain schemas, exclusion rules, and boundary enforcement rules without scheduling, approval, or runtime intermediation. The modify right per A2.02 applies: content-domain specifications can be updated through governed directed selection (B1.14) that creates new DNA versions (B2.69). Rule authoring per A2.04 is how content-domain specifications are created: they are authored as DNA-layer content, subject to the same authoring discipline as all governed substrate content. Without B1.06's DNA layer with its governance affordance applicability, content-domain specifications cannot receive the full governance treatment Paper 1's affordances provide. The result would be content-domain specifications that exist but cannot be governed — a fragile condition where the entity's operational territory is specified but not subject to the human authority that makes the specification defensible.

B1.18 provides DNA layer operational scope. The DNA layer per B1.06 contains behavior rules; B1.18 provides the operational scope that gives those rules their territory. Without B1.18, the DNA layer contains behavior specifications for an entity whose scope is not governed. This produces a specific architectural gap: the entity's behavior is governed but its territory is not, which means the boundary at which governance begins is not itself governed. Content-domain specifications resolve this gap by making the operational boundary a DNA-layer commitment — governed through the same infrastructure that governs everything else in the DNA layer.

The co-presence requirement:

B1.18 without B1.06 (content-domain without governed DNA layer). Content-domain exists as design documentation without governed substrate implementation. The specifications cannot be inspected per A2.01, modified per A2.02, or authored per A2.04 as architectural substrate. In the absence of the DNA layer, the entity may still exhibit something like content-domain behavior — refusing certain inputs, handling others — but the behavior’s source is not governed substrate content. This is the condition that can produce Implicit Content-Domain (B3.19): an entity that behaves as if it has content-domain without the authored, governed substrate specifications that make content-domain an architectural commitment.

B1.06 without B1.18 (DNA layer without content-domain). The DNA layer exists with behavior specifications but without operational territory definition. The entity’s behavior is governed through the DNA layer; its scope is not. Behavior rules apply when inputs arrive, but the DNA layer does not specify which inputs should arrive. Over time, this produces a system where the entity’s operational territory is implicitly defined by what inputs happen to reach it rather than explicitly authored as governed DNA-layer content.

Full co-presence. Content-domain specifications per B1.18 reside in the DNA layer per B1.06, governed through Paper 1 affordances (A2.01 inspect, A2.02 modify, A2.04 author), authoritative per A1.08, with boundary enforcement rules as operational DNA-layer content. The entity’s behavior is governed; its territory is governed; the boundary between in-domain and out-of-domain is governed substrate content, not implicit convention.

5. Interaction with Both Layers: Specification and Evidence

The enabling pair interacts with both layers of B1.06, not only the DNA layer. This two-layer interaction deserves explicit treatment because it clarifies what belongs in each layer and why.

DNA layer: content-domain specifications (governed policy). All four categories of content-domain specification identified in §2 reside in the DNA layer. These are governed policy content: authored, stable, subject to directed selection for modification, authoritative as the entity’s statement of operational territory. Changes to content-domain specifications proceed through DNA evolution mechanisms — directed selection per B1.14 producing new DNA versions per B2.69 — not through the Action layer.

Action layer: boundary violation events (operational evidence). When the entity encounters an out-of-domain input at runtime, that encounter is an operational event. The out-of-domain determination is made by applying the DNA-layer exclusion rules and boundary enforcement rules to the arriving input; the event record — that a particular input arrived, was determined to be out-of-domain, and was handled according to the DNA-layer out-of-domain handling rules — is Action-layer content per B2.26 and A2.40. The Action layer records content-domain boundary events as operational evidence.

This operational evidence has a governed path back to the DNA layer. Action-feedback evolution per B1.15 can draw on Action-layer boundary violation records to evaluate whether the entity's content-domain specifications are performing as intended: whether inputs are being correctly classified, whether exclusion rules are correctly calibrated, whether out-of-domain handling rules are producing appropriate responses. Where action-feedback analysis identifies content-domain refinement opportunities — inclusion rules too broad, exclusion rules incorrectly catching in-scope inputs, boundary enforcement producing unexpected outcomes — those refinements are proposed as new DNA-layer content-domain specifications, proceeding through DNA evolution's directed selection mechanism. The loop closes through governed channels: Action-layer evidence informs DNA-layer refinement through human-mediated action-feedback, without the Action layer directly rewriting DNA-layer content.

The architectural significance of this two-layer interaction is that content-domain governance operates at two distinct moments: at authoring time (DNA-layer specifications authored per A2.04 as the entity's operational territory) and at refinement time (Action-layer boundary evidence feeding DNA-layer specification updates through action-feedback per B1.15). Both moments are governed; neither is automatic.

6. Failure Modes

Two failure modes follow directly from absent or misplaced content-domain specifications in relation to the DNA layer.

Implicit Content-Domain (B3.19). This failure mode occurs when B1.18 content-domain specifications are absent from the DNA layer — when the entity exhibits content-domain behavior without governed substrate specifications. The entity accepts some inputs and refuses others; the basis for that discrimination is not authored DNA-layer content but accumulated practice, application logic, or emergent behavior. Implicit content-domain cannot be governed through Paper 1 affordances: there is no content-domain specification to inspect per A2.01, modify per A2.02, or version through directed selection per B1.14. The entity's operational territory is effectively ungoverned — determined by what happens to reach it and how it happens to respond, rather than by authored specifications that humans hold authority over. Detection: B2.90 content-domain specification requirements — are content-domain specifications present in the DNA layer? If not, or if the specifications present are documentation rather than governed substrate content, Implicit Content-Domain is present.

Layer Conflation (B3.07). This failure mode occurs when content-domain specifications are placed in the Action layer rather than the DNA layer. The placement may appear expedient — the Action layer is operational substrate content, content-domain boundary decisions produce operational records, and an implementer might conclude that content-domain specifications belong where boundary decisions are recorded. The placement is architecturally incorrect. Content-domain specifications are governed policy, not operational records. Policy content in the Action layer is not subject to the same

directed-selection modification pathway as DNA-layer content; it does not carry the same Category 4 authoritative status per A2.46; and it conflates two layers whose distinct governance semantics are a foundational commitment of the architecture. Detection: A5.10 source-of-truth test — are content-domain specifications correctly categorized as Category 4 authoritative DNA content? If content-domain specifications are found as Action-layer records rather than DNA-layer specifications, Layer Conflation is present.

7. Operational Detection and Pair Completeness

The pair B1.18 + B1.06 is fully co-present if and only if all of the following are true:

1. **B2.90 content-domain specification requirements met.** Content-domain specifications are present in the DNA layer: input domain schemas, exclusion rules, out-of-domain handling rules, and boundary enforcement rules are authored DNA-layer content, not documentation or application configuration.
2. **A5.10 source-of-truth test passed.** Content-domain specifications are correctly categorized as Category 4 authoritative DNA content per A2.46. A human exercising the inspect right per A2.01 can reach the specifications directly through the DNA layer without reconstruction from operational behavior or inference from other substrate content.
3. **B2.93 content-domain verification satisfied.** Specifications meet completeness requirements: the entity's in-domain territory is specified (not merely the out-of-domain territory), boundary enforcement rules are present and operational, and out-of-domain handling rules specify a governed response rather than an unspecified default.
4. **Governance affordance applicability confirmed.** Content-domain specifications are governed through the DNA layer's A2.01–A2.04 affordances. A human with appropriate access can inspect the specifications (A2.01), modify them (A2.02), and the specifications were or can be authored per A2.04. Modification proceeds through directed selection per B1.14 producing new DNA versions.
5. **Action layer boundary records present and distinct.** The Action layer contains boundary violation event records per A2.40 when boundary events occur. These records are distinct from the DNA-layer specifications — they are operational evidence, not policy content — and the path from Action-layer evidence to DNA-layer refinement runs through action-feedback per B1.15 with human mediation.

A system that passes all five tests has content-domain specifications that are authored, governed, authoritative, and supported by operational evidence through governed channels. A system that fails any test has a gap in the pair: either content-domain specifications are absent, mislocated, ungovernable, or the boundary-event feedback path is missing.

8. Conclusion

Content-domain per B1.18 and the two-layer structure per B1.06 are an enabling pair because each provides to the other a property neither can supply alone. B1.06's DNA layer gives B1.18's content-domain specifications a governed, A1.08-authoritative substrate home — the home that transforms content-domain from design intent to architectural commitment subject to Paper 1's governance affordances. B1.18's content-domain gives B1.06's DNA layer the operational territory specification that completes the DNA layer's governance of the entity: behavior rules without territory specification govern how without governing what, a gap that content-domain's authored specifications resolve.

The pair's interaction with both B1.06 layers — DNA layer as specification home, Action layer as boundary-event evidence repository — is not incidental to the architecture but a precise expression of the DNA/action distinction's governance semantics. Policy content belongs in the DNA layer; operational evidence belongs in the Action layer; refinement flows from evidence to policy through governed action-feedback channels, not through direct Action-to-DNA writes. Any deployment that violates this assignment — through implicit content-domain specifications or through Layer Conflation — has degraded one or both of the pair's foundational commitments in a way that cannot be recovered without restoring the co-presence condition.

Source Papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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